For Examiner's Use

4 A satellite of mass 1.0 kg orbits the Earth in a circular path.

Figure 4.1 shows the gravitational potential energy of the satellite at a distance r from the centre of the Earth in the region above the surface of the Earth. R_E is the radius of the Earth, which is 6400 km.

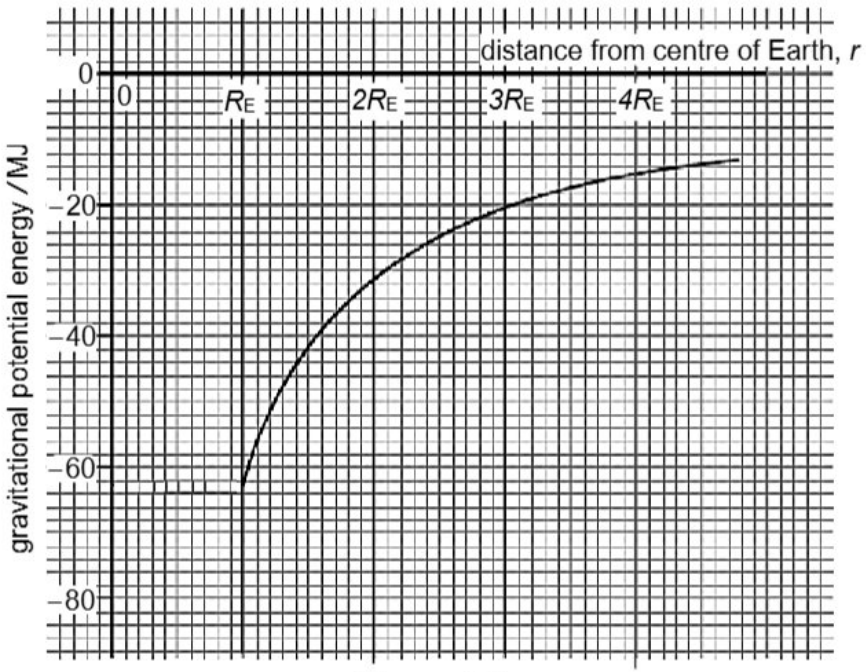


Figure 4.1

(a) (i) Define *gravitational potential* at a point.

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.....

.....

..... [2]

(ii) Using Figure 4.1, deduce the gravitational potential at the Earth's surface.

gravitational potential = ... MJ kg^{-1} [1]

(b) (i) If a satellite of mass m is in an orbit of radius R around the Earth, show that its kinetic energy E_K is given by

$$E_K = \frac{GMm}{2R}$$

where M is the mass of the Earth and G is the gravitational constant.

[2]

(ii) On Figure 4.1, sketch the graph of the variation of the total energy E_T of the satellite of mass 1.0 kg with r for values of r greater than R_E.

Explain clearly how you obtained the graph.

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..... [3]

(c) (i) Write down an equation which relates gravitational force F_G, and gravitational potential energy U.

..... [1]

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(ii) Using Figure 4.1, determine the gravitational force experienced by the satellite at a distance $2R_E$ from the centre of the Earth.

gravitational force = ... N [2]

[Total: 11]